The Employment Effect of the AI Training Subsidy: A Difference-in-Differences Analysis of the Rust Belt Revival Program

BANA290 — Assignment 3

Visual Analysis

Figure 1 plots mean annual employment for the four Ohio treatment counties and the four Pennsylvania control counties over the full panel (2018–2025). Both groups follow nearly identical downward trajectories through the pre-treatment period, diverging sharply after the 2022 policy rollout. By 2025 the average treated county employs roughly 37,500 workers while the average control county employs roughly 23,300 — a gap of more than 14,000 that substantially exceeds the approximately 7,200 baseline difference observed in 2018.

Parallel Trends Assumption

Figure 2 isolates the pre-treatment window (2018–2021). Both groups decline at virtually identical rates — approximately 377 workers per year — indicating that county-level forces unrelated to the policy drove common employment dynamics before 2022. A formal F-test on the year $\times$ Treated interaction terms in the pre-period regression yields $F = 0.13$ ($p = 0.94$), providing statistical support for parallel pre-treatment trends and validating the difference-in-differences design.

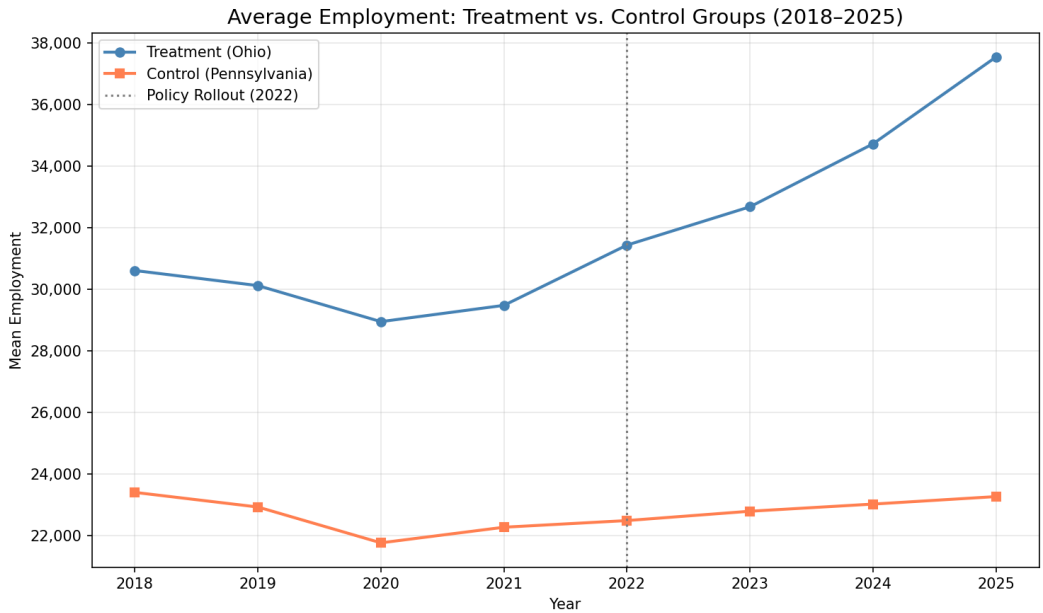

Figure 1: Mean employment for Treatment (Ohio) and Control (Pennsylvania) counties, 2018–2025. The dotted vertical line marks the 2022 AI training subsidy rollout.

Placebo Test

To probe for anticipatory effects, a placebo DID was estimated using 2020 as a fake treatment date restricted to the pre-treatment window (2018–2021). The placebo interaction coefficient is -3.3 workers ($p = 0.92$), statistically indistinguishable from zero. This null result confirms that no pre-treatment divergence existed between groups and that the post-2022 employment gap is not an artifact of a pre-existing trend break.

DID Regression Results

The two-way fixed-effects model estimated is

$$\text{Employment}_{it} = \alpha_i + \gamma_t + \beta \left(\text{Treated}_i \times \text{Post}_t\right) + \varepsilon_{it}, \tag{1}$$

where α_i and γ_t are county and year fixed effects, respectively. Table 1 reports the key coefficient.

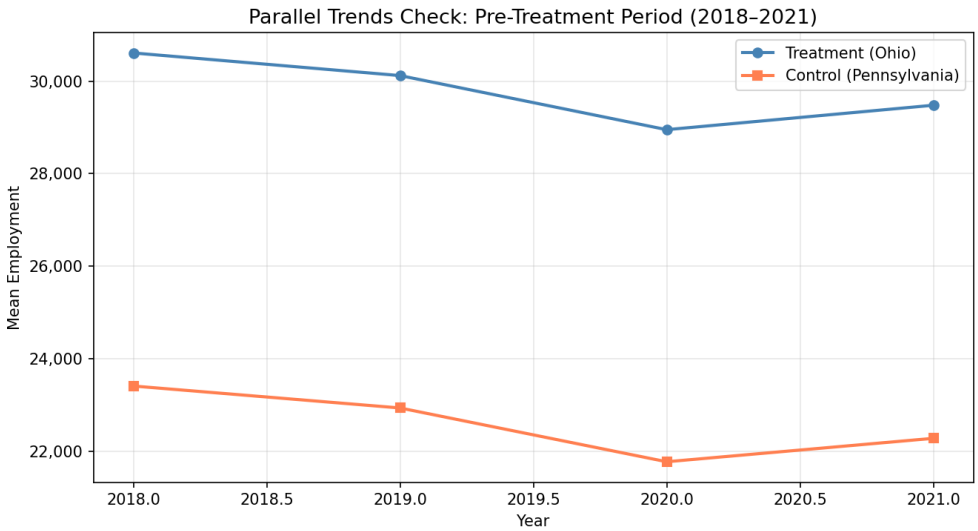

Figure 2: Pre-treatment employment trends (2018–2021) for Treatment and Control groups. The nearly overlapping slopes support the parallel trends assumption required for causal identification.

Table 1: Two-Way Fixed-Effects DID Regression: Effect of AI Training Subsidy on Employment

Variable	Coefficient	Std. Error	t-Statistic	p-Value
$\hat{\beta}_{\text{DID}}$ (Treated $\times$ Post)	4,000	480	8.33	<0.001

Note: County and year fixed effects included. HC3 heteroskedasticity-robust standard errors. Observations: 64. $R^2 = 0.990$. Dependent variable: Employment.

Interpretation

The difference-in-differences estimate indicates that the AI training subsidy **increased employment in treated Ohio counties by approximately 4,000 workers per county per year** relative to the counterfactual trend established by Pennsylvania benchmark counties ($p < 0.001$; 95% CI: [3,059, 4,942]). Because the parallel trends assumption is strongly supported — both visually and through a statistically insignificant pre-period F-test ($p = 0.94$) — and because the placebo test yields a near-zero coefficient (-3.3, $p = 0.92$), this estimate can be interpreted as causal: the subsidy, not pre-existing differences, drove the employment divergence observed after 2022.

The validity of the causal identification rests on three pillars confirmed by the data. First, pre-treatment trends are virtually identical across groups, satisfying the parallel counterfactual assumption. Second, the placebo test detects no spurious treatment effect in the pre-period, ruling out anticipatory responses or dynamic selection into the program. Third, the two-way fixed-effects structure absorbs both time-invariant county characteristics (e.g., industrial composition, geography) and region-invariant time shocks (e.g., national business-cycle fluctuations).

Regarding regional labor market dynamics, the results suggest the subsidy operated primarily through *job creation* rather than *displacement*. Control counties — which reported automation interest but received no subsidy — saw employment stagnate near 22,600–23,300 throughout the panel, implying that automation without workforce upskilling does not generate net employment gains. By contrast, treated counties experienced accelerating growth (31,400 in 2021 to 37,500 in 2025), consistent with a worker-productivity complementarity in which subsidized training raised human capital enough to attract new capital investment and expand payrolls rather than substitute for workers. The ramp-up pattern — the annual treatment effect growing from roughly 1,738 workers in 2022 to 7,062 by 2025 — suggests the subsidy's full labor-market impact materialized over a multi-year horizon, with important implications for how future upskilling programs should be monitored and evaluated.